1 June 1971
SUPERSEDING
MIL-M-43530
16 August 1967

MILITARY SPECIFICATION

MEASURES, LIQUID; PLASTIC, GRADUATED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for one type and three sizes of plastic liquid measure, equipped with a loop handle.

1.2 Classification.- The plastic liquid measure shall be furnished in the following sizes, as specified (see 6.2).

Size 1 - 1/2 pint
Size 2 - 1 pint
Size 3 - 1 quart

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-394	-	Plastic Molding Material, (Propylene Plastics, Injection and Extrusion).
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.

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MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-22748 - Plastic, Polyethylene and Copolymers, High Density, Molding and Extrusion Material.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Food and Drug Administration, Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

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3. REQUIREMENTS

3.1 Commercial product.- The plastic measure shall be the manufacturer's standard commercial design, except as specified herein.

3.2 Materials.- The materials used in the manufacture of the item shall conform to the requirements specified herein.

3.2.1 Plastic.-

3.2.1.1 Resin purity.- The linear polyethylene or polypropylene used in the manufacture of the measure and the antioxidants or other substances incorporated in the plastic shall conform to the Federal Food, Drug and Cosmetic Act, Food additives, Section 121.2501, subpart F, Part 121, chapter 1, title 21, Code of Federal Regulations for contact with food.

3.2.1.2 Linear polyethylene.- The measure and loop handle shall be made from virgin polyethylene material conforming to class A, grades 1, 3, or 5 of MIL-P-22748 with a density range of 0.941 to 0.985. Clean, unburned, plastic material in the form of imperfect parts, sprues, runners, or other scrap of the same composition as the virgin material may be reground and mixed with the material specified.

3.2.1.3 Polypropylene.- The measure and loop handle shall be made from virgin polypropylene material conforming to type I of L-P-394. Clean, unburned plastic material in the form of imperfect parts, spruces, runners, or other scrap of the same composition as the virgin material may be reground and mixed with the material specified.

3.3 Design and construction.- The plastic measure shall be of the pitcher type, having a pouring spout and loop-type handle on the side opposite the pouring spout. It shall be constructed in one piece from the material specified in 3.2.1.2 or in 3.2.1.3. The measure shall be constructed to hold its rated capacity while standing upright on a level surface. The capacity shall be as specified (see 6.2) and shall be graduated in the increments, shown in table I. The graduation marks shall be easily seen and numbered at the top mark (which shall be below the pouring spout) and at three or more intermediate points. Capacities shall be legibly and permanently embossed on each size measure. The measure and handle shall be sufficiently rigid so as to show no visible deflections or distortion when tested as specified in 4.2.3.

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TABLE I.- Size and increments

Size	Increments
1/2 pint	1 oz.
1 pint	1 oz.
1 quart	2 oz.

3.4 Volume accuracy.- The volume reading of the finished plastic measure shall be accurate within ± 5 percent when tested as specified in 4.2.3.

3.5 Leakage.- The finished plastic measure shall show no evidence of leakage when tested as specified in 4.2.3.

3.6 Marking for identification.- The outside bottom of the finished plastic measure shall be marked with the manufacturer's name or trademark and the date of manufacture.

3.7 Color.- The color of the measure shall be natural.

3.8 Finish.- The surfaces of the finished plastic measure shall be smooth, free from blisters, mold marks, weld marks, holes, cracks, lumps, and other defects that will affect the appearance or utility of the item.

3.9 Workmanship.- The finished plastic measure shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.1.1 Certificates of compliance.- Components and materials listed below may be accepted on the basis of a supplier's certificate of compliance for requirements specified in applicable paragraphs of this specification.

Component	Characteristic	Requirement paragraph
Plastic material	Purity	3.2.1.1
Plastic material	Composition	3.2.1.2
Plastic material	Composition	3.2.1.3

Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2.2 Examination of the end item.- The end item shall be examined in accordance with the classification of defects in 4.2.2.1. Random samples shall be drawn from each lot of end items for examination of visual defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of one measure each for the examination in 4.2.2.1. The sample unit for examination in 4.2.2.1 shall be one finished plastic measure. The inspection level shall be I and the acceptable quality level (AQL) shall be 2.5 defects per 100 units.

4.2.2.1 Visual examination.- The finished plastic measure shall be examined for defects in material, design, construction, appearance, workmanship, and marking.

Examine	Defect
Color	Not as specified. Discolored, not uniform in color, faded.
Construction and workmanship	Rim, handle, pouring spout and bottom not free of fins or flash, rough or sharp edges. Flash at other points exceeds 1/32 inch in length. Not smooth and uniform in texture. Warped, twisted, malformed, crazed, pitted, porous or bubbled; heat mark, cold seam, blister or uneven surface.

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Examine	Defect
Construction and workmanship (cont'd)	Measure and handle not of single piece construction. Handle not loop type. Handle twisted or malformed. Graduation marks missing. Graduation marks illegible or not located as specified. Any hole, crack, break, rupture, or fracture.
Appearance	Oil, dirt, or foreign matter. Chalked, burned, bloomed, or scaly. Any scratch, dent or nick on outside of measure affecting appearance.
Marking for identification	Manufacturer's name or trademark missing from bottom of measure.

4.2.2.2 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Markings (exterior and interior)	Incorrect, incomplete, illegible; omitted; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of case liner.
Contents	Number of measures per unit package is more or less than required. Number of unit packages per shipping container is more or less than required.

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4.2.2.2.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-601 or PPP-B-636, examination for defects in closure, waterproofing, and reinforcing shall be in accordance with the applicable specification.

4.2.3 Testing of the end item.- Testing of the completely fabricated plastic measure shall be performed in accordance with table II for the characteristics shown therein. The sample unit shall be two finished measures. The inspection sampling plan shall be in accordance with the following:

Lot size (lot unit)	Sample size (sample unit)
800 or less	2
801 up to including 22,000	3
22,001 and more	5

The lot size shall be expressed in units of one measure each. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

TABLE II.- Testing of the end item

Characteristic	Requirement paragraph	Test method	Requirements applicable to sample unit	Number of determinations per sample unit	Results reported as
Rigidity	3.3	4.3.1	X	1	Pass or fail
Volume accuracy	3.4	4.3.2	X	1	Pass or fail
Leakage	3.5	4.3.3	X	1	Pass or fail

4.3 Test methods.-

4.3.1 Rigidity.- The finished plastic measure shall be filled with water at 110° F to the top graduation mark, then raised and the water poured out. The measure and handle shall be examined for visible deflections or distortions, during or after the operation.

4.3.2 Volume accuracy.- The volume reading, when the finished measure is standing upright on its level base, shall be as specified in 3.4 and when filled to the top mark with water at 70° F from a calibrated graduate.

4.3.3 Leakage.- The finished measure shall be filled to the top graduation mark with water at 70° F and allowed to stand upright on a level surface for 24 hours. At the end of this period it shall be examined for evidence of leakage.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.- Six measures, of one size only arranged in two rows of three each, shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF (variety SW) class domestic, grade 200 of PPP-B-636. The measures shall be separated by individual half-slotted partitions made of the same material as the box. The inside depth of the box and height of the half-slotted partitions shall be 1/4 inch greater than the height of the measures. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C.- Measures shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.- Seventy-two size 1 measures, forty-eight size 2 measures, or thirty-six size 3 measures, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V2s, of PPP-B-636; or overseas type, style A, grade A or B, type 2 load of PPP-B-601, as specified (see 6.2). Each fiberboard container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Each wood container shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547; and closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B.- Seventy-two size 1 measures, forty-eight size 2 measures, or thirty-six size 3 measures, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each fiberboard shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.2.3 Level C.- Measures, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at

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the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The plastic measure is intended for domestic and institutional use for measuring liquids where precise measurements are not required.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Type of shipping container required for level A packing (see 5.2.1).
(e) When weather-resistant grade V3c or V3s fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Classification comparison.- The 1250 milliliter measure covered by MIL-M-43530 has been deleted from this revision since there are no known requirements. The 1/2 pint, 1 pint, and 1 quart measures have been added to replace metal items of equal capacity.

Custodians:

Army - GL
Navy - MC
Air Force - 84

Preparing activity:

Army - GL

Project No. 7240-0169

Review activities:

Army - MD
Navy - SA

User activities:

Navy - CG, YD

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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